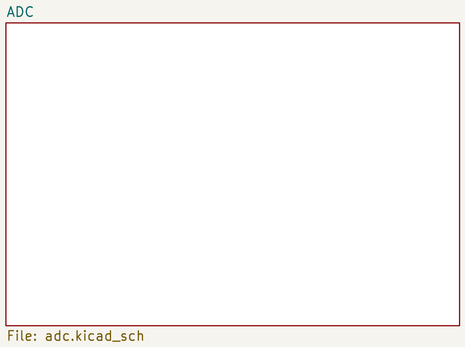
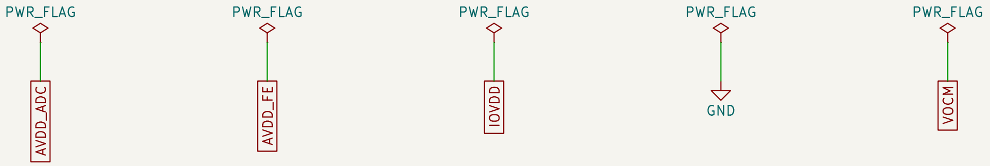


Root sheet. Vimal adds power.kicad_sch and test.kicad_sch here as more sheets. The PWR_FLAGs below are TEMPORARY so ERC passes before the power sheet exists: delete them once the LDO outputs drive AVDD_ADC, AVDD_FE and IOVDD



IEEE UIUC EOH 2027
Sheet: /
File: adc_board.kicad_sch

Title: EOH ADC board – root

Size: A4
Date: 2026-10-26
Rev: 0.1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/2

ADC SHEET – EOH lung array ADC board (owner: Tommy). 2x TLV320ADC5140 + 2x THS4524 (quad) front end + DB25 input + BRK8305 MC6/MC9 plug

Generated from gen_adc_sch.py (netlabel style). Pins connect by the label at the end of each stub. Rails (AVDD_ADC, AVDD_FE, IOVDD), PWR_GOOD and test points come from Vimal's power sheet.

Source of values: project-map/adc-board-circuit.md (front end, decoupling), fpga-interface.md (I2C addresses, TDM, pins). Tables: kicad/adc-sheet-tables.md

INPUT J1: DB25 female, TASCAM pinout

UMC1820 8 TRS line outs -> DB25-to-8xTRS snake

Shell ground-lift: R12 fitted 0R; if hum, remove R12 and fit R13+C23

CH1 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC A (U1) IN1

Amplifier: U4 section 1 (VIN1+/− , VOUT1+/−), on the quad THS4524 below

R109/R110 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

CH2 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC A (U1) IN2

Amplifier: U4 section 2 (VIN2+/− , VOUT2+/−), on the quad THS4524 below

R209/R210 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

CH3 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC A (U1) IN3

Amplifier: U4 section 3 (VIN3+/− , VOUT3+/−), on the quad THS4524 below

R309/R310 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

CH4 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC A (U1) IN4

Amplifier: U4 section 4 (VIN4+/− , VOUT4+/−), on the quad THS4524 below

R409/R410 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

VOCM: 1.474 V from AVDD_ADC divider, buffered by TLV9061 (matches ADC5140 internal input bias ~1.48 V). DNP R11 to let each THS4524 VOCM float to 1.65 V

CH5 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC B (U2) IN1

Amplifier: U5 section 1 (VIN1+/− , VOUT1+/−), on the quad THS4524 below

R509/R510 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

CH6 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC B (U2) IN2

Amplifier: U5 section 2 (VIN2+/− , VOUT2+/−), on the quad THS4524 below

R609/R610 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

CH7 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC B (U2) IN3

Amplifier: U5 section 3 (VIN3+/− , VOUT3+/−), on the quad THS4524 below

R709/R710 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

CH8 front end: THS4524 diff MFB low-pass, gain 0.2 (−14 dB), 50 kHz Butterworth, DC-coupled to ADC B (U2) IN4

Amplifier: U5 section 4 (VIN4+/− , VOUT4+/−), on the quad THS4524 below

R809/R810 = Cc position (1206): 0R = DC-coupled (default). AC fallback: fit 22uF and set CHx_CFG0 = 0x84

FRONT-END AMPLIFIERS: 2x THS4524 quad fully differential amp (U4 = CH1–4, U5 = CH5–8). Same amplifier as THS4521, four per package.

Every PD pin tied to AVDD_FE (always on). All VS− pins to GND. Each VSn+ pin has its own 100n (C_08 of that channel), each VOCMn pin its own 100n (C_07).

ADC A: TLV320ADC5140 (U1), I2C 0x4C, channels 1–4 -> TDM slots 0–3

U1 decoupling: each cap right at its pin.
VREF/MICBIAS cap grounds go straight to AVSS pin 4, no via.
Confirm values vs datasheet Fig. 172 / EVM schematic.

ADC B: TLV320ADC5140 (U2), I2C 0x4D, channels 5–8 -> TDM slots 4–7

U2 decoupling: each cap right at its pin.
VREF/MICBIAS cap grounds go straight to AVSS pin 4, no via.
Confirm values vs datasheet Fig. 172 / EVM schematic.

FPGA PLUG J2/J3: 2x20 2.00 mm socket onto BRK8305 MC6

MC6 = banks 85/86 at 1.8 V, no ground pins. J3 1x20 socket onto MC9 = all GND (ground return).

All signals on bank 86. Unused MC6 pins are left open (they are FPGA I/O: never tie them to GND).
Check on the real BRK8305 which MC6 row (odd or even) is nearer MC9; the clock pins (BCLK 18, FSYNC 22, SDOUT 26) are o

MC6–1 = B86_L9N
MC6–3 = B86_L9P
MC6–5 = B86_L10N
MC6–7 = B86_L10P
MC6–9 = B86_L11N
MC6–11 = B86_L11P
MC6–13 = B86_L12N
MC6–15 = B86_L12P
MC6–17 = B85_L12N
MC6–19 = B85_L12P
MC6–21 = B85_L11N
MC6–23 = B85_L11P
MC6–25 = B85_L10N
MC6–27 = B85_L10P
MC6–29 = B85_L9N
MC6–31 = B85_L9P
MC6–33 = B85_L8N_HDGC
MC6–35 = B85_L8P_HDGC
MC6–37 = B85_L7N_HDGC
MC6–39 = B85_L7P_HDGC

MC6–2 = NC
MC6–4 = NC
MC6–6 = NC
MC6–8 = B86_L1N
MC6–10 = B86_L1P
MC6–12 = B86_L2N
MC6–14 = B86_L2P
MC6–16 = B86_L7N_HDGC
MC6–18 = B86_L7P_HDGC
MC6–20 = B86_L8N_HDGC
MC6–22 = B86_L8P_HDGC
MC6–24 = B86_L3N
MC6–26 = B86_L3P
MC6–28 = B86_L4N
MC6–30 = B86_L4P
MC6–32 = B86_L5N_HDGC
MC6–34 = B86_L5P_HDGC
MC6–36 = B86_L6N_HDGC
MC6–38 = B86_L6P_HDGC
MC6–40 = B85_L6N_HDGC

DIGITAL GLUE: I2C pull-ups, SHDNZ pull-down, 33R series, SDOUT_B link

R6 fitted: both ADCs share SDOUT (TI scheme). R6 removed: ADC B uses its own lane SDOUT_B (MC6–30).
R4/R5 sit at each ADC's SDOUT pin; R7/R8 sit at the MC6 socket.

Owner: Tommy. Generated by gen_adc_sch.py
IEEE UIUC EOH 2027
Sheet: /ADC/
File: adc.kicad_sch
Title: ADC board – ADC sheet
Size: A1
Date: 2026-10-26
Rev: 0.1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 2/2